Assignment 05

Answer 01

Given, Total test instances = 45

Error rate = 6.67 %.

Confidence level = 95 %.

Z-value = 1.96

$$\text{Error Rate} = \frac{6.67}{100} = 0.0667$$

$$\text{Standard error} = \sqrt{(0.0667 \times 0.9333 / 45)}$$

$$= 0.0372.$$

$$\text{Margin of error} = 1.96 \times 0.0372$$

$$= 0.0729.$$

$$\text{Lower limit} = 0.0667 - 0.0729 = -0.0062$$

$$\text{Upper limit} = 0.0667 + 0.0729 = 0.1396$$

Since, the error rate cannot be negative, we take the lower limit as 0.

So, The 95 % confidence interval for the true error rate is approx 0 % to 13.96 %.